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PROPOSAL 1: DRMC Outpatient Department Patient Flow

A. Project Title and Description

Title: Simulation of Patient Flow in the Outpatient Department of Davao Regional Medical Center, Apokon, Tagum City Using Discrete Event Simulation

Description: This project simulates the flow of patients through the Outpatient Department (OPD) of Davao Regional Medical Center (DRMC), located at Apokon Road, Brgy. Apokon, Tagum City, Davao del Norte. DRMC is a Level III Department of Health teaching-training hospital. Its authorized bed capacity was increased by Republic Act No. 11879 from 600 beds to 1,000 beds, indicating its role as a large public referral medical center for Tagum City, Davao del Norte, and nearby service areas.

This proposal focuses on the scheduled, walk-in, follow-up, and referred outpatient consultation processes within the DRMC OPD, which serves between 500 and 1,000 patients per day on typical weekdays. The OPD operates Monday to Friday from 8:00 AM to 5:00 PM. It utilizes a combination of online appointment-based scheduling and walk-in/referred processing.

The simulation will model the step-by-step journey of OPD patients starting from early morning arrival (including pre-opening queues forming at 5:00 AM or earlier), gate screening, appointment-code/referral-slip verification, baseline vital signs collection, central registration (document checking and encoding), clinic routing, waiting room queues, doctor consultations, secondary diagnostic/laboratory lines, post-consultation social services (such as Malasakit funding), pharmacy queues, follow-up scheduling, and final exit.

The purpose is to create a highly realistic operational model of the DRMC OPD patient-flow process using Discrete Event Simulation (DES). This model will allow administrators to identify bottlenecks and test structural scheduling or staffing adjustments without interfering with actual daily clinical operations. The project is strictly designed to analyze operational flow, queue lengths, waiting times, resource utilization, and patient throughput; it does not simulate clinical diagnoses or medical outcomes.

B. Problem Definition

Davao Regional Medical Center operates in Tagum City, which had a 2024 population of 300,042, while Davao del Norte had a 2020 Census population of 1,125,057. As a government regional medical center and Level III teaching-training hospital, DRMC may receive outpatient consultation demand from Tagum City, nearby municipalities, follow-up patients, referred patients, and patients seeking specialty services. This makes its OPD a realistic setting for studying appointment-based patient-flow congestion.

The specific operational problem is that OPD patients experience extreme delays because early morning surges, high walk-in and referred demand, document verification, vital sign checks, and clinical consultations are handled under constrained resource levels.

The specific DRMC OPD bottlenecks to be modeled include:

1. **The Dawn-to-Opening Bottleneck (Pre-Opening & Front-End Registration Logjam):**
Patients traveling from distant provinces arrive at the facility as early as 3:00 AM or 4:00 AM to secure consultation slots. This results in a massive, highly congested pre-opening queue that peaks between 7:00 AM and 10:00 AM (especially on Mondays). When the registration and triage desks open, the front-end processes, managed by only 1 to 2 verification staff and 2 to 3 central registration encoders, are immediately overwhelmed by hundreds of patients simultaneously checking PhilHealth documents, capturing baseline vital signs, and sorting new versus returning records.
2. **The "Slowed" Sub-Clinic Consultation Line:**
The absolute longest wait occurs at the face-to-face consultation stage inside the sub-clinics, with the Internal Medicine (IM), Obstetrics and Gynecology (OB-GYN), and Pediatrics clinics driving the primary congestion. Because DRMC is a training hospital, resident doctors (30 to 45 in the morning, 20 to 35 in the afternoon) are frequently pulled away for emergency room or ward duties. This results in an **interruption event** that causes a **partial staffing reduction** where an estimate of approximately 75% of the scheduled doctor staffing remains active in the clinic, while the remaining 25% of capacity is temporarily reassigned. This reduced capacity state lasts for 3 to 5 hours without active updates, significantly slowing patient flow, compounding wait times, and causing physical discomfort in overcrowded hallways.
3. **Ancillary and Post-Consultation Bottlenecks:**
After completing consultation, patients do not immediately exit. Instead, they face secondary lines: 40% to 60% of patients are routed to diagnostic services (laboratory, imaging, ECG, ultrasound, CT scan), and many must navigate social service funding queues (such as the Malasakit center) and the outpatient pharmacy to secure subsidized government medications.

This simulation aims to pinpoint exactly where and why these queues form, and to provide hospital administration with a risk-free platform to test operational modifications (such as increasing staff, adjusting triage priority rules, and isolating clinical consultation lines from emergency ward duties).

C. Objectives of the Simulation

1. **Map and Model the Real-World Patient Journey:** Build an operational model mapping the DRMC OPD flow from early morning arrival (5:00 AM queue start), through gate screening, verification/vitals, central registration, sub-clinic waiting, doctor consultation, ancillary diagnostics/pharmacy, and final exit.
2. **Isolate and Quantify Major Bottlenecks:** Measure the precise wait times associated with the front-end triage desks, sub-clinic consultation lines (specifically accounting for doctor **staffing reductions** of 25%, maintaining 75% capacity, due to ward/ER interruptions lasting 3 to 5 hours), and secondary post-consultation lines.
3. **Incorporate and Evaluate Priority Lane Dynamics:** Evaluate the performance and impact of the dedicated express lanes established across nearly all major stages (Entry & Triage, Registration & PhilHealth, Consultation nursing rotation, and Lab/X-Ray/Pharmacy) for vulnerable sectors (Senior Citizens, PWDs, and pregnant patients).

4. **Measure Key Performance Indicators (KPIs):** Collect baseline data across standard and specialty paths, utilizing the specific metrics outlined in the KPI table below.

5. **Test Alternative Intervention Scenarios:**

- *Scenario A (Front-End Scaling):* Increase registration staff from the current 2 to 3 encoders to the recommended 5 to 6 staff, and verification staff from 1 to 2 up to 3 to 4 staff.
- *Scenario B (Consultation Staggering & Caps):* Model a strict scheduling cap of 15 to 20 patients per session for resident physicians, and test the impact of scheduling doctor sessions without ward/ER double-duty interruptions (i.e., maintaining 100% doctor staffing capacity throughout the session).
- *Scenario C (No-Show Optimization):* Simulate backfilling the estimated 10% to 20% no-show appointment slots with waiting walk-in or referred patients.
- *Scenario D (Mid-Week Training Staffing Shifts):* Model the staffing drops that occur during mid-week (Wednesday/Thursday) medical trainings and month-end resident physician rotation handovers.

Main KPIs to be produced by the model:

KPI	Meaning in the DRMC OPD Simulation
Pre-Opening Queue Build-up	Average number of patients waiting outside the entrance lobby prior to the 8:00 AM official opening (simulated starting at 5:00 AM).
Average Gate & Verification Waiting Time	Time from patient arrival at the perimeter or entrance queue to the completion of appointment-code/referral-slip verification and baseline vitals capture.
Average Central Registration Wait Time	Time spent queuing to submit PhilHealth/ID documents and encode clinical data (distinguishing between new and returning patients).
Clinic Corridor Queue Wait Time	Waiting time in sub-clinic corridors (e.g., Internal Medicine, OB-GYN, Pediatrics) before being called for doctor consultation.
Duration of Sub-Clinic Consultation Staffing Reductions	Total simulated duration where sub-clinic doctor staffing is reduced to approximately 75% capacity due to resident doctors being called to ER or ward duties.
Average Consultation Service Time	Active face-to-face time spent with the consulting physician.
Average Post-Consultation Line Delays	Cumulative time spent in secondary queues for diagnostics, Malasakit social service billing, and pharmacy dispensing.
Average Total OPD Length of Stay (LoS)	Total time elapsed from initial arrival (including pre-opening arrival) to final exit.

Resource Utilization Rates	Percentage of time verification staff, registration staff, and consulting doctors are active versus idle.
Patient Throughput	Total number of patients fully processed per sub-clinic per day.

D. Identified Entities, Events, and Resources

Entities:

- **Patients:** The primary dynamic entities moving through the system, defined by their priority status, registration type, and clinical path:
 - *Priority Status:* General Patients versus Priority Patients (Senior Citizens, PWDs, pregnant patients, and infants).
 - *Registration Type:* New Patients (requiring full digital profiling) versus Returning Patients (requiring chart retrieval).
 - *Routing Source:* Scheduled Appointment (25%), Follow-up (35%), Walk-in (20%), or Referred (20%).
- **Diagnostic Requests:** Secondary entities generated for the 40% - 60% of patients requiring laboratory, imaging, or specialized diagnostic tests.
- **Follow-up Requests:** Secondary entities representing patients scheduled for future appointments.

Events:

- **Pre-Opening Arrival:** Initiation of the patient queue at 5:00 AM at the outer perimeter.
- **Gate Screening & Line Sorting:** Sorting of arrivals into walk-in, online appointment, and priority lanes.
- **Verification & Vitals Check:** Capturing baseline physiological data and verifying booking codes or referral slips.
- **Central Registration Intake:** Inputting new patient details or retrieving existing medical charts.
- **Sub-Clinic Check-in:** Dropping routing slips into clinic boxes (e.g., Internal Medicine, Pediatrics, OB-GYN).
- **Emergency Pull-out / Doctor Staffing Interruption:** A random or schedule-driven event where active doctor resources are temporarily reduced by 25% (leaving 75% of active staffing in the sub-clinic) due to ward/ER duty calls.
- **Doctor Consultation:** Active clinical assessment.
- **Ancillary Routing:** Sending patients to secondary lines (diagnostics, billing, pharmacy).
- **Follow-up Booking:** Recording future return appointments.
- **Patient Exit:** Exit of patient from the OPD system.

Resources:

- **Gate Security & Entry Staff:** Stationed at the outer perimeter to guide patients into correct queues.
- **OPD Verification Staff:** Current: 1 to 2 staff; Optimized Scenario: 3 to 4 staff.
- **Health Aides:** Stationed at the entrance lobby desk to capture vital signs.
- **Administrative Registration Staff (Encoders):** Current: 2 to 3 staff; Optimized Scenario: 5 to 6 staff.
- **Clinic Routing Staff / Entrance Triage Desk:** 1 to 2 staff routing unassigned patients.

- **Sub-Clinic Nursing Staff:** Manage sub-clinic check-in boxes and administer the priority rotation queue.
- **OPD Physicians (Resident and Specialist Doctors):**
 - Morning Session: 30 to 45 doctors pooled across all specialties.
 - Afternoon Session: 20 to 35 doctors pooled across all specialties.
 - *Resident Cap:* Limit of 15 to 20 patients per doctor per session.
- **Sub-Clinic Consultation Rooms:** Physical clinic spaces.
- **Diagnostic Channels & Pharmacy Counters:** Service channels representing lab, imaging, billing, and pharmacy counters.

E. Operational Modeling Assumptions

The following parameters are derived directly from the DRMC OPD operational questionnaire that was answered by an employee at DRMC Outpatient Department, not the supervisor. These values serve as the baseline for the simulation model.

1. **Operating Hours and Cycle:** The simulation models a standard 24-hour cycle, with the active OPD operational window from 8:00 AM to 5:00 PM, Monday through Friday. Front-end queues begin accumulating in the model starting at 5:00 AM.
2. **Patient Volume:** Daily outpatient volume ranges between 500 and 1,000 consultations per day, modeled as a pooled Poisson arrival process.
3. **Arrival Distribution & Peak Dynamics:**
 - *Dawn Surge (5:00 AM – 7:00 AM):* High arrival rate representing patients traveling from distant provinces.
 - *Peak Queue Congestion (7:00 AM – 10:00 AM):* Maximum arrival rate of approximately 60 to 80 patients/hour on non-Mondays, and up to 100 to 120 patients/hour during the Monday morning surge.
 - *Midday Stabilization (10:00 AM – 1:00 PM):* 30 to 45 patients/hour.
 - *Afternoon Decline (1:00 PM – 5:00 PM):* 15 to 25 patients/hour.
4. **Patient Mix (Source Distribution):**
 - Scheduled Online Appointment: 25%
 - Follow-up: 35%
 - Walk-in: 20%
 - Referred (from external facilities): 20%
5. **Clinic Routing Distribution:**
 - Internal Medicine (IM): 25% - 30%
 - Pediatrics: 15% - 20%
 - Obstetrics/Gynecology (OB-GYN): 15% - 18%
 - Surgery: 10% - 12%
 - Orthopedics: 6% - 8%
 - Ophthalmology: 4% - 6%
 - Pediatric Immunization: 3% - 5%
 - Dental: 2% - 4%
 - Dermatology: 2% - 3%

- Unassigned/Needs Routing: 5% - 7%
6. **Front-End Processing Times (Triangular Distributions):**
- *Appointment Code/Referral Verification:*
 Triangular ($a = 0.5$, $b = 2.5$, $c = 15$) minutes
 (Min: 30s – 1min; Typical: 2 - 3min; Max: 10 – 15min due to document discrepancies)
 - *New Patient Registration (Full Profile Creation):*
 Triangular ($a = 7$, $b = 17.5$, $c = 60$) minutes
 (Min: 7 – 10min; Typical: 15 – 20min; Max: 45 – 60 min during network/document delays)
 - *Returning Patient Registration (Chart Retrieval & Update):*
 Triangular ($a = 1$, $b = 4$, $c = 20$) minutes
 (Min: 1 – 2min; Typical: 3 – 5min; Max: 15 – 20min during physical chart search)
7. **Clinical Consultation Parameters:**
- *Doctor Staffing Levels:* Morning: 30 - 45 doctors; Afternoon: 20 - 35 doctors.
 - *Patient Cap:* Resident physicians are capped at seeing 15 - 20 patients per session.
 - *Consultation Service Time:* Triangular ($a = 5$, $b = 15$, $c = 45$) minutes
 (Min: 5min for quick checkups; Average: 15min; Max: 45min for complex teaching-hospital clinical evaluations)
 - *Doctor Ward/ER Interruption (Staffing Reductions):* Modeled as a scheduled or stochastic capacity reduction event where doctor resources are temporarily reduced by 25% (meaning approximately 75% of normal scheduled staffing remains active in the clinic) with a frequency of 1 to 2 times/day per resident queue, lasting for a duration of:
 Triangular (180, 240, 300) minutes (3 – 5 hours)
8. **Priority Lane Multi-Stage Queuing:**
- Priority patients (Seniors, PWDs, pregnant patients) are modeled using a non-preemptive priority queuing rule (Priority = 1, General = 2).
 - *Verification & Vitals:* Access to 1 dedicated express window.
 - *Central Registration:* Access to 1 dedicated processing window.
 - *Consultation Queue:* Nursing staff use a 1:3 rotation rule (slotting 1 priority patient ahead of every 3 general patients in the doctor's queue).
 - *Diagnostics & Pharmacy:* Dedicated priority windows bypass standard queues.
9. **No-Show & Late Arrival Handling:**
- The no-show rate is set at 10% - 20%. Unused appointment slots are dynamically reallocated to waiting walk-in or referred patients.
 - Late-arriving scheduled patients (who arrive after the clinic has reached capacity or after their session has ended) are routed to a "Rescheduling/Discharge" state or asked to return on another day.
10. **Consultation Outcomes and Ancillary Routing:**
- *Direct Exit (Prescription & Instructions only):* 45% of patients.
 - *Ancillary Diagnostics Routing:* 25% of patients.
 - *Direct Follow-up Scheduling:* 20% of patients.
 - *Departmental Referral / Direct Admission:* 10% of patients.

- *Aggregate Diagnostic Demand:* Across all clinics, 40% of consultations result in diagnostic orders.
- *Aggregate Follow-up Demand:* 65% - 75% of all seen outpatients require a future follow-up appointment.

F. Data Collection and Validation Requirements

To ensure the simulation model achieves high statistical validity and accurately reflects DRMC's specific operational patterns, the project team will focus on verifying the following target datasets:

Needed Data	Practical Collection Method	Validation Metric
Hourly Arrival Rates	Extract data from the electronic appointment queue system and gate logbooks.	Mean and variance of hourly arrival distributions.
Ancillary Turnaround Times (Diagnostics/Pharmacy)	Collect sample timestamps from the Laboratory Information System (LIS) and Pharmacy dispensing logs.	Average diagnostic turnaround times (40% - load).
Clinical Interruption Frequency	Conduct short-interval interviews with OPD Nursing Supervisors to record ward/ER call-outs.	Mean Time Between Interruptions (MTBI) and Mean Time to Repair (MTTR / Interruption Duration with 75% staffing retention).
Training Session Calendars	Coordinate with clinical department heads to obtain schedules for Wednesday/Thursday medical reviews and transition dates.	Programmed staffing capacity drops.

Source notes for factual grounding:

- **Davao Regional Medical Center Charter & RA 11879:** Establishes DRMC's Level III teaching-training classification, location, and its expanded 1,000-bed capacity.
- **DRMC OPD Appointment System & Portals:** Confirms the patient screening processes, requirements for PhilHealth/ID verification, and division of clinics.
- **Philippine Statistics Authority (PSA):** Confirms local demand context with Tagum City (300,042 population) and Davao del Norte (1,125,057 population) demographics.